Differentiation: Implicit Differentiation

Questions 1–30

Question 1. Find

a) $\frac{d}{dx}(x + y^2)$
b) $\frac{d}{dx}(xy)$
c) $\frac{d}{dx}(x^2y)$
d) $\frac{d}{dx}(e^x \sin y)$
e) $\frac{d}{dx}(e^y \cos x)$
f) $\frac{d}{dx}(x \ln y - 5)$
g) $\frac{d}{dx}(2^x y + 3)$

Question 2. For each of the following, find $\frac{dy}{dx}$:

a) $y^2 = 8x$
b) $xy = 12$
c) $x^3 + xy^2 - y^2 = 0$
d) $\ln x + y^2 = 2x$
e) $5x^2 - 2xy + 5y^2 = 25$
f) $\sin x + xe^y = 1$
g) $x \cos y - 5 = x$
h) $x \ln y + y \ln x = 10$

Question 3. By using implicit differentiation, find the tangent line to the circle $x^2 + y^2 = 25$ at the point $(4, 3)$.

Question 4. Use implicit differentiation on $e^y = x$ to show that $\frac{d}{dx}(\ln x) = \frac{1}{x}$.

Question 5. Use implicit differentiation on $\ln y = \ln a^x$ to show that $\frac{d}{dx}(a^x) = a^x \ln a$.

Question 6. Differentiation of Inverse Trigonometric Functions Use implicit differentiation on $\sin y = x$ to show that $\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$.

Question 7. Use implicit differentiation on $\cos y = x$ to show that $\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$.

Question 8. Use implicit differentiation on $\tan y = x$ to show that $\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$.

Question 9. Mix Find the value(s) of $\frac{dy}{dx}$ at the given point:

a) $\sin 2y - x = 2$ at $(-2, \pi)$

b) $x^2 y = x$ when $x = 2$

c) $xe^y - x^2 + 4 = 0$ at $y = \ln 3$

Question 10. Show that if $x^2 \sin y = 4$ then $\frac{dy}{dx} = -\frac{8}{x\sqrt{x^4-16}}$.

Question 11. Show that if $x^2 \tan y = 1$ then $\frac{dy}{dx} = -\frac{2x}{x^4+1}$.

Question 12. The equation of a curve is given by $x^2 + 2y^3 = 2$.

a) Find the coordinates of the stationary points on the curve.

b) Find the coordinates of the points where the curve has an infinite gradient.

Question 13. Prerequisite knowledge for Q13: Inverse Functions

a) Sketch the graphs of $y = f(x) = x^2 + 4$ for $x \geq 0$ and its inverse $y = g(x)$ on the same axes, labelling any intersections with the axes.

b) Show that $f'(3) = \frac{1}{g'(13)}$.

Question 14. Given that $f(x) = x(x-3)$, $x \geq 3$, calculate the gradient of the graph $y = f^{-1}(x)$ at the point $(10, 5)$.

Question 15. The equation of a curve is given by $(x-2)^2+2=e^y$. Find the coordinates of the point where the tangent line is parallel to the x-axis.

Question 16. Show that the curve $xy^2+\sqrt{y}=5$, $y>0$ has no stationary points.

Question 17. A curve has equation $(2x + y)^2 = x - y$.

a) Find the tangent lines to the curve at $x = \frac{1}{4}$.

b) Hence, show that the two tangent lines meet at the point $(-\frac{5}{12}, \frac{1}{3})$.

c) Find the exact area of the triangle bound by the two tangent lines and the x-axis.

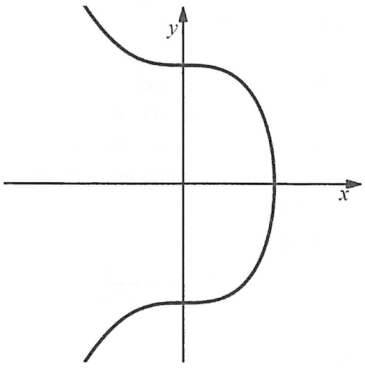
Question 18. Show that for the curve $5x^2 + 12xy - 4y^2 = 10$ there are no tangent lines parallel to the line $y = \frac{3}{2}x$.

Question 19. A curve has equation $(x+y)^3 = 10x^2 - 2y$.

a) Show that the point $(1,1)$ lies on the curve.

b) Show that the tangent to the curve at the point $(1,1)$ is given by $7y - 4x = 3$.

Question 20. The curve $x^3 + y^2 = 5$ is shown in the diagram below.



a) Show that the point $(1,2)$ lies on the curve.

b) Find the normal line to the curve at the point $(1,2)$ in the form $ax + by = c$ where a, b and c are integers to be found.

c) Show that the normal line found in part b) does not intersect the equation again.

Question 21. The equation of a curve is given by $x^2 \sin y + \cos y = \frac{1}{2}x$.

a) Show that $(0, \frac{3\pi}{2})$ lies on the curve.

b) Show that $\frac{dy}{dx} = \frac{4x \sin y - 1}{2 \sin y - 2x^2 \cos y}$.

c) Find the equation of the tangent to the curve at the point $(0, \frac{3\pi}{2})$ giving your answer in the form $y = mx + c$.

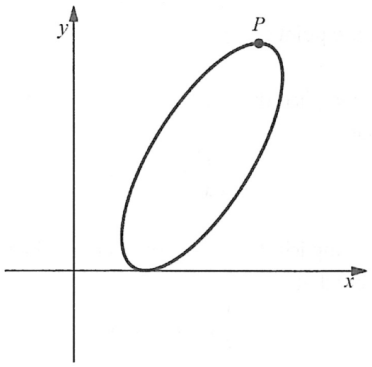
Question 22. The equation of a curve is given by $x^2 - 2x \ln y = 1$.

a) Show that $(1, 1)$ lies on the curve.

b) Show that $\frac{dy}{dx} = \frac{y(x - \ln y)}{x}$.

c) Find the equation of the normal to the curve at the point $(1, 1)$ giving your answer in the form $y = mx + c$.

Question 23. The curve $2(x-5)^2 - 2xy + (y+1)^2 = 1$ is shown in the diagram below.



a) Find the coordinates of the maximum point P of the ellipse.
b) Show that the distance between where the ellipse meets the x-axis and the point P is $k\sqrt{5}$ where k is an integer to be found.

Question 24. The curve $y^2 + \ln x = k$, $x > 0$ where $k > 0$ passes through the point $(1, -\sqrt{2})$.
a) Find the value of k.
b) Find the exact coordinates at which the tangent to the curve is parallel to the y-axis.

Question 25. The equation of a curve is given by $xe^y = 32$.

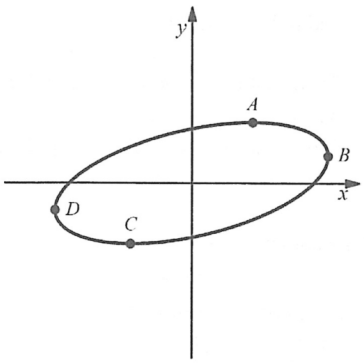
a) Show that the point $(2, 4\ln 2)$ lies on the curve.

b) Find the equation of the tangent line and the normal line at the point $(2, 4\ln 2)$.

c) Find the values where the equation of the tangent line and the normal line meet the x-axis.

d) Hence show that the triangle bounded by the tangent and normal line at $(2, 4\ln 2)$ and the x-axis has an area of $20(\ln 2)^2$.

Question 26. Prerequisite knowledge for Q26: Domain and Range of Functions A curve is given by $x^2 - 2xy + 5y^2 = 20$ and shown on the diagram below.



The points A, B, C and D lie on the curve. The domain is such that the curve lies between the x-values of D and B. The range is such that the curve lies between the y-values of C and A. Find the coordinates of A, B, C and D.

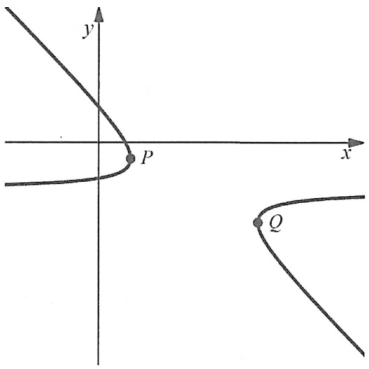
Question 27. A curve is given by $xy^2 + x^3y = k$ where $k \neq 0$.

a) Show that the curve never crosses the x-axis.

b) Show that $\frac{dy}{dx} = \frac{-y^2-3x^2y}{2xy+x^3}$.

c) Given that a stationary point occurs when $x = -1$, find the value of k.

Question 28. A curve has equation $3x + xy + y^2 = k$ where k is a positive integer such that $k < 9$.



There are two points $P(a, b)$ and $Q(5a, 5b)$ on the curve such that the tangents to the curve are parallel to the y-axis.

a) Find the values of a, b and k.

b) State the coordinates of the points P and Q.

Question 29. A curve has equation $(x+y)^4 = 2x - y + 8$.

a) Show that the point $(-2, 3)$ lies on the curve.

b) Find the tangent line to the curve at the point $(-2, 3)$ in the form $ax + by = c$ where a, b and c are integers to be found.

c) Prove that the tangent line found in part b) does not intersect the equation again.

Question 30. Prerequisite knowledge for Q30: Newton-Raphson Method The percentage of woodland cover in the UK, P, since 1965 may be modelled by the equation

$$P = 8.5 + \arctan(t - 31)$$

where t is the number of years since 1965.

a) Show that $\frac{d}{dx}(\arctan x) = \frac{1}{x^2+1}$.

b) Hence, find $\frac{dP}{dt}$.

c) Use Newton-Raphson's method starting at $t_1 = 30$ to find the year in which, according to the model, the percentage of woodland coverage is 9.95%.